National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

Programming Fundamentals (CS1002) — Mid-2 Exam

Date: November 8th 2025

Course Instructor(s)

Rizwan Ul Haq, Dr. Rabia Maqsood, Tahir Farooq, Umer Iqbal, Mughees Ismail

Total Time (Hrs.): **1 hour**
Total Marks: **45**
Total Questions: **3**

Vetted by: _____ **Signature:**_____

Roll No.: _____ Section: _____ Student Signature: _____

	Q-1	Q-2	Q-3	Total
Total Marks	15	15	15	**45**
Marks Obtained				

Attempt all the questions on this sheet.

Q1	CLO 2	15 Marks

An airport ticketing system calculates airfare based on the **travel class** and **age category** of the passenger.

Rules:

1. **Travel class** (entered as a character):
   - E → Economy
   - B → Business
   - F → First Class
2. **Base fares**:
   - Economy → 300
   - Business → 500
   - First Class → 800
3. **Age-based discount** (on the base fare):
   - Age < 12 → 50% discount (child)
   - Age between 12 and 60 → no discount (adult)
   - Age > 60 → 30% discount (senior citizen)
4. **Special Rule:**
   - If the passenger is **above 80**, add an **extra 10% service fee** after the discount.

Write a C++ program that:

- Takes two inputs:
  1. travelClass (E/B/F)
  2. age
- Calculates and displays the **final ticket price** after applying all conditions.
- Displays **"Invalid class or age input"** if the class character or age is invalid (e.g., negative value in age or class values are other than E, B, F).

  Sample input:
  Enter Travel Class: B
  Enter Age: 85
  Sample Output:
  Final Ticket Price: 385

```
#include <iostream>
using namespace std;

int main() {
    char travelClass;
    int age;
    double baseFare = 0.0, finalFare = 0.0;

    cout << "Enter Travel Class (E/B/F): ";
    cin >> travelClass;

    cout << "Enter Age: ";
    cin >> age;

    // Validate inputs
    if (age < 0 || (travelClass != 'E' && travelClass != 'B' && travelClass != 'F'))
{
        cout << "Invalid class or age input" << endl;
        return 0;
    }

    // Assign base fare
    switch (travelClass) {
        case 'E': baseFare = 300; break;
        case 'B': baseFare = 500; break;
        case 'F': baseFare = 800; break;
    }

    // Apply age-based discount
    if (age < 12)
        finalFare = baseFare * 0.5; // 50% discount
    else if (age <= 60)
        finalFare = baseFare;       // no discount
    else
        finalFare = baseFare * 0.7; // 30% discount

    // Apply extra service fee if above 80
    if (age > 80)
        finalFare = finalFare * 1.10; // add 10% service fee

    cout << "Final Ticket Price: " << finalFare << endl;

    return 0;
}
```

Criteria	Description	Marks
1. Input Handling & Validation	Correctly takes travel class (E/B/F) and age as inputs; validates both.	**3 marks**
2. Base Fare Assignment (Class logic)	Correctly assigns base fare according to class (E=300, B=500, F=800).	**2 marks**
3. Age-based Discount Logic	Correct discount applied (50% for <12, none for 12– 60, 30% for >60).	**4 marks**
4. Special Rule (Age > 80)	Applies additional 10% service fee **after** discount for ages above 80.	**3 marks**
5. Output Formatting & Clarity	Displays correct final ticket price or appropriate error message.	**2 marks**
6. Code Style & Structure	Proper indentation, meaningful variable names, clean structure.	**1 mark**

Q2	CLO 2	15 Marks

A *"number transformer"* machine that changes a number every day based on special rules:

i. If the number is even, it becomes half of itself.
ii. If the number is odd, it becomes 3 × number + 1.
iii. After each transformation, the sum of its digits is subtracted from the new number.
iv. The process continues until the number becomes 1 or less.

Write a C++ program using **while loops only** to simulate this process and count how many transformations occur before the number reaches 1 or below. Take the input from the user.

Sample input:
n=28
Sample output:
Step 1: 28 → after transformation: 9
Step 2: 9 → after transformation: 18
Step 3: 18 → after transformation: 0
Final number: 0
Total transformations: 3

```
#include <iostream>
using namespace std;
int main() {
    int number;
    cout << "Enter a positive integer: ";
    cin >> number;
    int transformations = 0;
    while (number > 1) {
         int before = number;
        if (number % 2 == 0)
            number = number / 2;
        else
            number = 3 * number + 1;

        //Find sum of digits using while loop
        int temp = number;
        int digitSum = 0;
        while (temp > 0) {
            digitSum += temp % 10;
            temp /= 10;
        }
        //Subtract digit sum
        number = number - digitSum;
        transformations++;
        cout << "Step " << transformations << ": " << before << " ? after
transformation: " << number    << endl;
    }
    cout << "Final number: " << number << endl;
    cout << "Total transformations: " << transformations << endl;

    return 0;
}
```

Q3	CLO 3	5*3 = 15 Marks

For the given C++ code snippets, write the output (avoid overwriting, your answers should be clear).

```
int x = 7, y = 2;
double result;

do{
   cout<<"\nCurrent values: x = " << x << ", y = " << y;
   result = (x % y == 1)
            ? static_cast<double>(x) / y
            : x / static_cast<double>(y + 1);
   cout<<"\nResult = " << result;

   x -= 2;
   y++;
}while (x > y);

cout<<"\nFinal values: x = "<<x<<", y = "<<y;
```

Your Output:

```
Current values: x = 7, y = 2
Result = 3.5 //(1.5pts)
Current values: x = 5, y = 3. //(1pt)
Result = 1.25  //(1.5pts)
Final values: x = 3, y = 4. //(1pt)
```

```
int code = 3, marks = 8;
int count = 1;

while(count <= 3) {
   switch (code) {
       case 1:
           cout<<"Coding ";
           marks += 5;
           break;
       case 2:
           cout<<"is ";
           marks *= 2;
       case 3:
           marks = marks / 2 + 7 * count;
           cout<<"FUN ";
       default:
           cout<<"isn't ";
           marks -= 3;
   }
   code = (marks > 15) ? 2 : (code < 10 ? 1 : 3);
```

```
    count++;
}
cout<<"\nFinal Marks: "<<marks;
```

Your Output:

```
FUN isn't Coding Coding  //(0.5+1+1+1 pts)
Final Marks: 18  //(1.5pts)
```

```
int i, total = 0, items = 3;
double val;

for(i = 1; i < 4; i++){
   val = items-- * i;
   cout<<"I have got " << val;

   if(val > 3.0 && total){
       cout<<" candies";
       total += items;
   }
   else if (val < 5.0)
       if(!items){
          cout<<" croissants";
          total += (i * 3) % 4;
       }
       else
          cout<<" bagels";
          total += i + static_cast<int>(val / 2);
   cout<<"\nTotal bill = "<<total<<endl;
}
```

Your Output:

```
I have got 3 bagels. //(0.5+0.5 pts)
Total bill = 2 //(1pt)
I have got 4 candies //(0.25+0.25 pts)
Total bill = 7 //(1pt)
I have got 3 croissants  //(0.25+0.25 pts)
Total bill = 12 //(1pt)
```

-0.5 for not considering newline or running loops for extra iterations